

MindfulBuddy: Extracting Comprehensive Breathing Biomarkers for Breathing Exercise Biofeedback Using Earbud Motion Sensors

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Abstract—Slow-paced deep breathing exercises have many health benefits, including stress management, lowering blood pressure, pain management, and controlling pulmonary conditions. While biofeedback can significantly improve the efficacy of breathing exercises, existing approaches support limited biomarkers, such as breathing rate, for specific breathing exercises (e.g., equal-phase breathing) in particular conditions without considering the breath-holding phase or variation in device orientation. Therefore, there needs to be a more convenient and robust approach that can generate and deliver comprehensive digital breathing biomarkers to facilitate biofeedback for various types of breathing exercises. In this article, we present a system with lightweight algorithms to passively track mindful breathing in real-time using lower-power earbud motion sensors to extract fine-grained comprehensive breathing biomarkers for generating biofeedback on users' breathing exercises. We utilize the earbud's motion sensor data to detect nonbreathing head motion and develop an extensive set of breathing markers, including breathing phases, breathing depth, breathing rate, breathing symmetry, and breath-holding. Such a comprehensive set of biomarkers can enable engaging user experience and effective mindful breathing exercises toward better stress management and overall mental well-being. Moreover, we develop a physiologically informed, novel earbud orientation handling algorithm that makes our biomarkers more resilient to ear canal shape and size. Finally, we showcase potential use-cases based on the breathing biomarkers derived from our algorithms to provide biofeedback on user's overall breathing performance.

Index Terms—Breath-hold, breathing, breathing depth, breathing phase, earbud mindfulness, hearable.

I. INTRODUCTION

MINDFUL, slow-paced, deep breathing exercises are scientifically studied strategies to reduce stress and pain, improve calmness, increase focus, and improve overall mental well-being [1], [2]. Studies show that deep breathing exercises decrease levels of salivary cortisol [3], [4], blood glucose [5], [6], as well as plasma rennin levels [7].

Studies suggest that slow-paced deep breathing has an immediate calming effect on the hypothalamus-pituitary axis (HPA) and autonomic nervous system (ANS) response to

stress. Sympathetic nervous system response, such as heart rate increases during inhalation and decreases during exhalation. This response is called respiratory sinus arrhythmia (RSA) – cardiac interbeat-interval (IBI) oscillations. Breathing directly modulates the RSA [8], [9]. Breathing exercises cause a shift toward parasympathetic nervous system dominance via direct vagal stimulation [10]. These findings support the idea that regular practicing of deep breathing helps reduce cumulative effects of stress in the body through improved affective processing [11], [12] and immune function [13], [14].

Biofeedback in breathing exercises helps an individual learn to control physiological responses by displaying in real-time how users' breathing affects measured physiological biomarkers [15]. Studies in athletes suggest that deep breathing with biofeedback is an effective, safe, and easy tool for both athletes and coaches to improve sports performance [16]. However, despite multiple pieces of evidence on the effectiveness of biofeedback, most breathing exercise-supporting apps and tools either depend on uncomfortable, expensive sensing devices (e.g., respiratory inductive plethysmography [17], [18] chestband) that are not ideal for daily use, or applicable to specific breathing patterns (e.g., equal phase breathing [19], [20]) and unable to generate critical breathing biomarkers (e.g., depth of breathing [19], [20], [21], [22] or breath-holding [20], [21]) for feedback on the actual breathing behavior and the exercise performance [1].

In this article, we present MindfulBuddy, an earbud-based passive sensing system that offers fine-grained comprehensive breathing biomarkers in real-time during slow-paced deep breathing exercises using low-power motion sensors. Earbuds are becoming increasingly popular consumer devices and embedding many sensing capabilities, including motion sensors. One of the biggest advantages of using earbuds for passive breathing monitoring is that the earbud is placed in a fixed location with a constant and close distance from the lung and the respiratory tract. Moreover, chest expansion and contraction due to breathing also move the head and earbuds synchronously. Our system utilizes this opportunity to passively monitor users' mindful breathing exercises and their quality to enable breathing biofeedback without additional expense. The major goal of MindfulBuddy is to provide a practical and convenient means to obtain comprehensive breathing biomarkers that can potentially be used to build numerous applications aiming to promote

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long-term practice of mindful breathing exercises and stress management. To the best of our knowledge, this is the first work showing the feasibility of tracking fine-grained, comprehensive breathing biomarkers associated with mindful breathing exercises, including breathing depth (shallow versus deep), real-time on-going inhalation and exhalation phases, breathing symmetry, and breath-holding all in one place for providing breathing biofeedback using consumer earbuds.

We handle the following key challenges in building MindfulBuddy.

- 1) Shallow breathing produces a very low magnitude range in the IMU motion sensor signal, which makes the signal difficult to distinguish between breathing versus absence of breathing (e.g., earbud on a table).
- 2) Nonbreathing head motion falls into the same frequency as the breathing head motion, which makes the traditional signal filtering algorithms fail in removing this motion artifact.
- 3) Earbuds inside the ear can have different orientations based on users' ear canal shape, size, and preference.
- 4) Breath-holding phases can show breath-like signals due to baseline shift or slow bodily motion.

We handled the above challenges in MindfulBuddy system by distinguishing shallow breathing from deep breathing, nonbreathing head motion from breathing head motion, and developing a physiologically informed algorithm for projecting earbud orientation to foot-to-head body orientation.

In summary, our contributions are the following.

- 1) We present an earbud motion sensor-based sensing system that can track breathing patterns and capture fine-grained breathing biomarkers, all using earbud motion sensors.
- 2) We present algorithms to estimate breathing depth with a 94.67% accuracy using motion sensors embedded in consumer earbuds and distinguish shallow breathing from deep breathing for generating breathing depth-related biofeedback.
- 3) We present algorithms to detect breath-hold with 95.02% accuracy during breathing exercises.
- 4) We present algorithms to track inhalation and exhalation phases with 89.5% accuracy in real-time using earbud gyroscope data to monitor various breathing exercises.
- 5) We present a novel, physiologically informed orientation handling algorithm that makes the accelerometer-based breathing exercise tracking independent of orientation.
- 6) We present an algorithm to detect nonbreathing head motion as a critical confounding motion artifact for earbuds and distinguish it from breathing head motion to generate biofeedback related to the head motion.
- 7) We present the evaluation of our algorithms, including a wide range of paced breathing exercises. We further implement our algorithms on-device and show the use cases that our algorithms can work in low-resource settings, such as earbuds.
- 8) We show the design of a practical earbud-based breathing biomarker sensing system that provides real-time

biofeedback about the breathing exercise and its confounders, such as motion artifacts and unexpected breathing patterns.

II. BACKGROUND AND RELATED WORKS

In this section, we review the literature on breathing biomarker monitoring algorithms and breathing biofeedback technologies utilizing smartphones and wearable devices. Moreover, we highlight the differences between our approach and the existing methods.

A. Biomarkers for Breathing Exercise Monitoring

Among other applications of mobile health [23], [24], [25], breathing biomarker detection is a challenging task and relatively new. Breathing is a complex physiological process of inhaling oxygen from the air and exhaling carbon dioxide from the body. The activity of regular breathing is an autonomous function performed without conscious effort. This type of breathing is usually shallow and quicker in nature [26]. It ranges between 12 and 20 breaths per minute (BPM) for healthy adults [27]. Conscious breathing, such as guided breathing exercises have been observed to affect various mental and physical well-being conditions positively. These type of breathing are deep and slow-paced ranging from 5 to 10 BPM [19], [20]. Breathing rate referred to the number of BPM is the most commonly used biomarkers to provide breathing biofeedback [20], [21], [22].

Based on the target impact of breathing rate on mental and physical well-being, trained care providers often guide individuals in different variations of breathing exercises, which generally entail breathing in (i.e., inhalation) for a specific time and breathing out (i.e., exhalation) after a particular time. Depending on the exercise, there might be another phase called breath holding after the inhalation and exhalation phases. Breath-holding refers to holding one's breath for a specific time, as suggested by the exercise.

Breathing exercises have varying intents. For example, equal breaths ('sama vittri'), coherence breathing [28] and box-breathing [29] (4-s inhale, 4-s hold, 4-s exhale, 4-s hold) are supposed to improve relaxation, breath of fire [30] is to increase calmness, ocean breath ('ujjayi') [31] is to empower focus, 4-7-8 breath [32] (4-s inhale, 7-s hold, 8-s exhale) is to reduce anxiety. The inhalation/exhalation (I/E) ratio also plays a vital role [33]: a short inhalation period followed by a long exhalation period is more effective at inducing relaxation than the opposite. In our article, we will refer to the ratio of inhalation/exhalation as breathing symmetry.

Some recent studies show the promise for capturing breathing phase and symmetry during slow-paced breathing exercises without the performance validation [20], [21]. Whereas, breathing depth is a critical biomarker for breathing exercises which indicates amount of air inhaled inside the lung during breathing. However, none of the existing works that offers breathing depth biomarker or detect breath-holding as a biofeedback to the user. Our system captures all the breathing

biomarkers, including breath-hold detection by utilizing earbud motion sensors.

B. Breathing Biomarker Estimation and Breathing Biofeedback

1) *Breathing Biomarker Estimation*: To capture different biomarkers related to breathing, a sensor or device next to the mouth or nose is a common approach to estimating airflow/breathing [37]. Breathing-related body movements can also be easily extracted by using inertial measurement units (IMUs) attached to chest bands [27], [38], head-mounted devices [39], and wrist-worn devices [40], [41]. Several of those works either focus on user-initiated breathing measurement [39] or work in constrained environments, such as sleep [41], [42]. Previous works also explored contact-less breathing measurement with a motion capture system [43], a smartphone microphone [44], and analysis of high-frequency wireless signals [45]. All previous approaches mentioned above show the feasibility of actively or passively detecting breathing using mobile sensors and focusing on breathing rate estimation. In this work, we focus on passive breathing phase detection, breathing rate, depth, and symmetry estimation in real-time using a low-power accelerometer embedded in commodity earbuds.

2) *Biofeedback Driven Breathing Exercises*: A class of smartphone and smart-wearable-based breath detection focused on controlled breathing to provide biofeedback for meditation [21], [22], [46]. Breeze [22] used smartphone acoustic data to detect breathing phases (inhale, exhale) to provide biofeedback in a particularly structured breathing exercise (4-s inhale, 2-s pause, 4-s inhale, 2-s pause with pursed-lip breathing). MindfulWatch [21] shows the feasibility of tracking breathing phases using smartwatch gyroscope data to generate biofeedback for a particular body posture and wrist position. However, they kept the real-time biofeedback as a future work. Another group of researchers worked on a smart necklace [46] that detects breathing patterns related to different affective valance and arousal to communicate the emotion to the users and their loved ones through audio, haptic, and visual representation. Moreover, JustBreath [8], CalmCommute [47], AmbientBreath [48] presented breathing biofeedback through haptic feedback embedded in a driver's seat or through virtual reality display or audio-visual guide to reduce driver's stress. These are one-way interactions where the breathing guidance is provided but do not assess whether the users follow the guidance.

3) *Commercial Breathing Monitors*: The consumer devices on the market that offer respiration monitoring can be divided into two categories based on their underlying sensing methodologies. Devices, such as Spire [49] and MonBaby [50] use the motion sensor to detect body movement associated with breathing. Spire tag needs to be attached to the waist-belt. On the other hand, MonBaby focuses on sleep breathing monitoring for infants, and the device needs to be attached to the baby's clothing around the chest. However, such products mainly provide a coarse respiration rate, which is insufficient

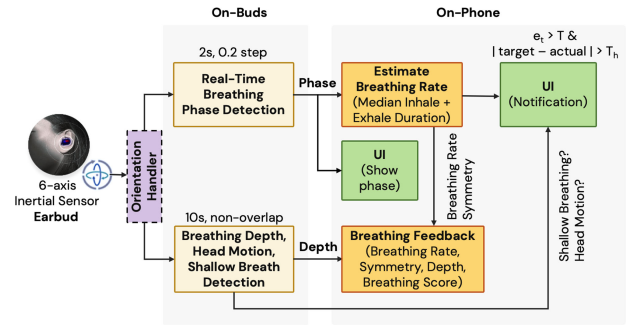


Fig. 1. MindfulBuddy system overview for passively tracking an on-going breathing exercise and extracting comprehensive breathing biomarkers to provide biofeedback. Here e_t is the elapsed time from the exercise onset, T is two times the breath durations, $target$ is the target breathing rate, $actual$ is the estimated current breathing rate and T_h is 2 BPM.

for meditation feedback. Polar Serene [19] offers biofeedback for slow-paced deep breathing exercises by utilizing photoplethysmography (PPG) signal based IBI tracking. It provides breathing rate and symmetry as the biofeedback. Another group of products, including Zephyr Bioharness [51] and Hexoskin [52] can be categorized as “wearable garment.” Similar to traditional respiratory inductive plethysmography, such devices exploit conductive material to measure breathing activities. However, these products are designed for monitoring breathing during exercise, such as running. For the application of breathing biofeedback, the proposed earbud system offers a much more convenient and cost-efficient solution.

4) *Earbud Sensor-Based Vital Sign Monitoring*: Recent works show the potential of earbud sensing to estimate breathing, heart rates, and heart rate variability [34], [53], [54], [55], [56], [57]. For example, inertial sensors available on earbuds, such as accelerometers and gyroscopes, are used to estimate breathing rate and heart rate [34], [53]. On the other hand, in-ear audio captured by the earbud's microphone is used to estimate both breathing rate [54], [55], [57], and heart rate [57]. Several of these works focus on a particular environment, such as sleep [54] or work with heavy breathing after exercise when the breathing sound becomes more audible [55]. These works establish the feasibility of monitoring both breathing and heart rates using the same earbud device. It shows the evidence that earbud is becoming popular sensing modality to estimate breathing and cardiac biomarkers. Our MindfulBuddy is pushing this effort to a new level for comprehensive breathing exercise monitoring in real-time using an earbud. Table I highlights the comparison between the breathing biomarkers offered by MindfulBuddy with the most relevant existing works in one place.

III. SYSTEM OVERVIEW

The proposed system MindfulBuddy (Fig. 1) uses a six-axis motion sensor (three-axis accelerometer and three-axis gyroscope) to extract breathing phase in real-time to guide user's breathing pattern. Fig. 1 shows the overview of the system.

TABLE I

COMPARISON BETWEEN OUR WORK WITH PRIOR WORKS USING COMMERCIAL DEVICES. OUR WORK PRESENTS THE MOST COMPREHENSIVE SET OF BREATHING BIOMARKERS FOR MINDFUL BREATHING EXERCISE TRACKING AND BIOFEEDBACK IN REAL-TIME INCLUDING NOVEL BREATHING PARAMETERS, SUCH AS BREATHING DEPTH, BREATH-HOLD, AND IE RATIO. HERE, ACC. STANDS FOR ACCELEROMETER, GYRO. STANDS FOR GYROSCOPE, MAE STANDS FOR MEAN ABSOLUTE ERROR, AND BPM STANDS FOR BREATH PER MINUTE

Prior Work	Type	Device	Sensor	Biomarkers	Subjects	Accuracy (Error)
Roddiger et al. [34]	Normal	Earbud	Accelerometer, Gyroscope	Breathing Rate	12	2.55 BPM (MAE)
RRMonitor [35]	Normal	Earbud	Accelerometer	Breathing Rate	30	2.33 BPM (MAE)
Ahmed et al. [26]	Normal	Earbud	Accelerometer, Audio	Breathing Rate	30	1.36 BPM (MAE)
Islam et al. [36]	Normal	Phone	Accelerometer, Audio	Breathing Phase Breathing Rate Inhale/Exhale ratio	131	77.3% 91.74% 86.76%
Hao et al. [21]	Deep	Watch	Gyroscope	Breathing Phase Breathing Rate	11	83% 0.77 BPM (MAE)
Polar Serene [19]	Deep	Watch	Photoplethysmograph (PPG)	Breathing Rate Inhale/Exhale Ratio	—	— —
Shih et al. [22]	Deep	Phone	Audio	Breathing Phase Breathing Rate	16	75.50% 2.27 BPM (MAE)
Rahman et al. [20]	Deep	Earbud	Accelerometer	Breathing Phase Breathing Rate	30	84.4%-86.94% 0.31 BPM (MAE)
MindfulBuddy	Deep	Earbud	Gyroscope Accelerometer, Gyroscope Gyro. Gyroscope Accelerometer	Breathing Phase Breathing Depth Breathing Rate Inhale/Exhale Ratio Breath-hold	30	89.5% 94.67% 0.61 BPM (MAE) 0.32 (MAE) 95.02%

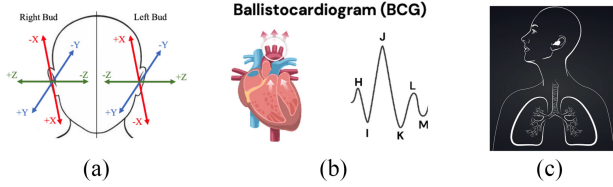


Fig. 2. (a) Baseline earbud orientation in the ear. Accelerometer has the following orientation when earbuds are worn – X: superior inferior, Y: anterior posterior, Z: lateral medial. The gyroscope 3-axis measures the rotational motion with respect to the X, Y, and Z axis, respectively. (b) Heart pumping the blood through arteries creates motion called ballistocardiogram (BCG) and corresponding BCG morphological signal complex called IJK is shown here. (c) Lung position in the body with respect to the earbud. Lung expansion and contraction due to inhalation or exhalation creates breathing motion which can be captured by the same set of earbud motion sensors.

- 1) **Physiologically Informed Orientation Handling:** Due to variation in ear canal shape and size of the users, earbud orientation inside the ear can vary user to user. Moreover, breathing motion during inhalation is opposite to the motion during exhalation. We need to determine the earbud orientation to correctly detect the breathing phase (inhalation versus exhalation). We develop a novel, physiologically informed earbud orientation handling algorithm that considers the strongest head motion due to heart pumping the blood (namely, ballistocardiogram (BCG) [58] response) to align the earbud orientation with foot-to-head body axis. Our algorithm selects the combination of accelerometer axes that capture the strongest BCG response across superior inferior and anterior posterior axis since breathing motion is also best captured through the same axis combination (Fig. 2). The detail of the orientation handling algorithm is explained in Section V-A.
- 2) **Real-time Breathing Phase Detection:** We use a 2-s long window with 200-ms sliding to detect the breathing

phase in real-time. We empirically determine the 2-s with 200 ms sliding window to have a minimum but enough data for capturing an ongoing breathing phase for real-time detection and feedback generation. Previous work [20] also used the same window size for tracking breathing exercises. Once a phase-change is detected, the breathing phase information and the corresponding timestamp are sent to the companion phone for biofeedback. The details of the phase detection algorithm are in Section V-C.

- 3) **Nonbreathing Head Motion, Shallow Breathing Detection and Breathing Depth Estimation:** We use a 10-s nonoverlapping window to calculate the depth, detect nonbreathing head motion and distinguish shallow breathing from deep breathing. The data segment assessing the breathing depth needs at least one inhalation or one exhalation phase. We design our algorithms to support slow-paced breathing as low as 3 BPM. It means that we need at least 10 s data to capture one phase. The head motion, shallow breathing status, and breathing depth information are sent to the connected phone for generating user feedback and further post-processing. The details of the depth calculation algorithm are described in Section V-B.

On the phone side, we use the phase information to show the user's current phase on the user interface (UI). If the user is inhaling, it shows 'You are inhaling,' and if the user is exhaling, it shows 'You are exhaling' on the UI. We use the timestamps to calculate the median inhale and exhale time from detected phases.

During the guided breathing exercise, if the user is not following the guidance, we use the breathing rate information to provide real-time feedback to the user. In the beginning, the user can select a target breathing rate by setting up the intended inhale and exhale duration for the session. If the estimated real-time breathing rate is greater than the threshold (T_h is

greater than 2 BPM [59]), we generate real-time feedback to the user to follow the guidance on the screen. We also generate a notification on the screen if the user is randomly moving their head or breathing is shallow.

Finally, when the user finishes the breathing exercise, we show the breathing biomarkers and a breathing performance score. The score indicates how closely the user followed the target breathing pattern. It is a way to objectively quantify users' breathing performance by providing feedback based on their actual breathing. Details of the breathing performance score is described in Section VII-A.

IV. BREATHING EXERCISE STUDY

We conducted an extensive breathing exercise-based data collection study with 30 participants (age range 23–54) in a 60-min lab study, where the participants performed 13 different breathing exercises. We recruited the participants using a consulting firm specializing in human factors and usability studies. Our legal and consulting firm's institutional review board (IRB) team approved the study. We discuss our data collection and annotation in the following.

A. Data Collection

We used Samsung Galaxy Buds2 paired with a Samsung Galaxy S9 smartphone for the data collection. The ground truth breathing signal is collected using a Zephyr Chestband Bioharness Biomodule3 [60].

1) *Sensor Data*: We collected 6-axis IMU data from Samsung earbuds at a 50 Hz sampling rate and the ground truth breathing waveform using Zephyr chestband at an 18 Hz sampling rate. Zephyr uses a pressure sensor to track breathing-related chest movement and is considered a reliable device for chest movement tracking [51], [61], [62]. We have also collected audio from the earbuds at 16 KHz sampling rate and electrocardiography (ECG) data from the chestband at 250 Hz. The sensor data was first stored in the paired phone and uploaded to a HIPAA-compliant [63] Amazon Web Storage (AWS) at the end of the data collection.

2) *Data Collection Protocol*: During the data collection, the participants were required to perform several natural and guided breathing sessions. At the beginning of the study, the participants performed six natural breathing sessions, where they breathed naturally in different postures (sitting, lying, and standing). Each breathing session lasted for 1 min. We chose 1 min inspired by previous breathing biofeedback development studies [20], [22]. After the natural breathing tasks, the participants performed guided breathing exercises with 3, 5, 10, 15, 20, 25, 30, 35, 40, and 45 breathing cycles (ten tasks). These breathing exercises can be considered 'equal breathing' as they have equal inhalation and exhalation duration. To assist the participants in performing the breathing exercise, we showed an animation on a phone app configured to show the breathing phase (inhale, exhale, or pause) instructions according to the number of breathing cycles. After the equal breathing exercise, participants performed nonbreathing head movement exercises, including slow and fast head movements, posture change, and random movements.

We collected head-motion data to develop our nonbreathing head-motion detection algorithm. Participants then performed three well-known breathing exercises with specific patterns: 1) 4-2-4-2 Breathing: 4-s inhale, 2-s hold, 4-s exhale, 2-s hold; 2) 3-4-5 Breathing: 3-s inhale, 4-s hold, 5-s exhale; and 3) 4-7-8 breathing: 4-s inhale, 7-s hold, 8-s exhale. Similar to equal breathing exercises, we showed the app animation to guide the participants. In addition, participants were also asked to hold their breath for as long as they could or up to a 1-min data for collecting data for breath-hold detection. Participants could take breaks between each breathing task to ensure their breathing rate returns to baseline and they are comfortable performing the next breathing task.

Finally, we test our demo breathing exercise tracking and the real-time feedback system using Galaxy Buds2 and an Android Smartphone. Participants performed a test breathing exercise with 5-s inhale and 5-s exhale (6 BPM) for 1 min (ten breathing cycles). We then asked our participants to participate in a usability survey using the standard System Usability Scale (SUS) [64]. Participants are also asked about the usability of the breathing quality score in understanding their breathing exercise performance and overall experience with the study.

A researcher administered the whole lab study and helped the participants follow the screen's guidance. The participants received \$200 for participating in the data collection study.

B. Data Annotation

We have developed a semi-automated process and an interactive interface for breathing data annotation, verification, and correction. This annotation interface is flexible enough to annotate single time-series data (e.g., chest-band breathing) and multiple time-series together (e.g., chest-band breathing and earbud motion sensor data in synchrony). It allows annotators to adjust the synchronization across different modalities so that each breathing cycle in the reference signal can be aligned with the earbud signal. We recruited a nurse as the breathing annotator with at least two years of experience performing breathing annotation. We first automatically annotate the peak and valley of the breathing cycles by using the algorithm described in [65]. The accuracy of this automated algorithm detects 3.1% spurious breathing cycles and misses 5.6% valid breathing cycles [65] for regular breathing data collected in a stationary condition. Error rate sometimes can be higher in case of breathing data affected by nonbreathing motion artifacts. However, the automated algorithm significantly reduces annotators' burden to annotate each breathing cycle.

An annotator can upload the raw breathing signal and corresponding automated annotations in our annotation interface to visualize, verify, and adjust the annotation. Furthermore, experienced breathing researchers verified the annotation performed by the nurse and further adjusted the annotation if any valid breathing cycle was missed or the spurious cycle was annotated as a valid one. We believe that several iterations of the data annotation, multimodal verification, and adjustment improve the reliability of the annotation.

C. Understanding Breathing Exercises and Performance

We analyze our study participants' breathing performance on the annotated data. The most common metric to compare the breathing performance is the mean absolute difference between the target breathing rate and actual breathing rate [34], [35], [59]. We show the user breathing rate performance for equal breathing and for breathing with breath-hold. In both cases, we observe that the slower the breathing rate, the higher the difference. Participants performed the best for breathing rates between 10 and 20 BPM. Breathing rates for healthy adults range from 10 to 20 BPM [35], [66]. As they perform slow-paced breathing, the difference gets bigger. We observe that the breathing rate target with 3 BPM (i.e., 10-s inhalation, 10-s exhalation) is the most challenging slow-paced breathing in our study, and the average difference is 10 BPM. Visual verification of the 3 BPM data reveals that several participants have taken stacked breathing and needed to catch up with their breaths while they were performing the relatively long inhalation (10 s) and exhalation (10 s). Hence, we observe a higher difference. It also indicates that not all slow-paced breathing exercise is suitable for everyone. We observe an increasing difference for breathing more than 35 BPM, as it is deviated more from the regular range.

V. ALGORITHMS TO ESTIMATE DIGITAL BIOMARKERS DURING BREATHING EXERCISES

In this section, we present lightweight algorithms on earbud motion sensor data to estimate comprehensive breathing biomarkers and assess users' breathing exercise performance quality. Breathing markers include breathing phases (inhalation, exhalation, hold/pause), breathing rate, depth, and symmetry (IE ratio). Finally, we present an example of computing users' breathing quality score by combining the common and user-interpretable breathing markers, such as breathing rate and depth.

A. Physiologically Informed Earbud Orientation Handling

Due to the variation of the earbud size and ear canal size, we can observe the variation of the earbud orientation from user to user. Moreover, tilted head positions can pose additional challenges to determining the earbud accelerometer orientation. We have developed a novel, physiologically informed orientation algorithm to handle these real-world challenges. Our algorithm aligns the earbud accelerometer axis with the foot-to-head axis of the earbud user. We determine the foot-to-head axis based on the strongest response of the heart motion, which is called Ballistocardiogram (BCG) [58]. Our algorithm detects J-peak using the most recent technique described in [67]. It first estimates IBI using Bruser's algorithm [68]. It then removes the IBI affected by motion artifacts and corrects some of the erroneous IBI by estimating instantaneous heart rate and considering a Gaussian prior. Finally, it detects the J-peak as the highest peak in the vicinity of each IBI estimate. Overall, it detects J-peaks with $95.84 \pm 1.90\%$ accuracy.

The highest BCG response are observed across the superior inferior axis followed by anterior posterior axis and lateral medial axis, respectively. Based on the J-peak magnitude from

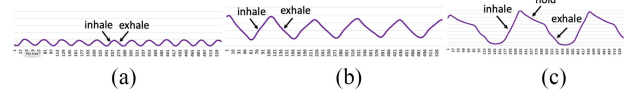


Fig. 3. (a) Shallow breathing. (b) Deep breathing without hold. (c) Deep breathing with breath-hold from the same subject. Each figure shows about 30 s of data on such breathing. Vertical axis shares the same scale and the normalized amplitude. Inhaling more air indicates deeper breathing. It means higher chest expansion is positively correlated with breathing depth and tidal volume [69].

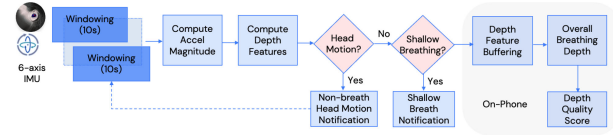


Fig. 4. Breathing depth determination algorithm steps.

X, Y, Z separately, our algorithm automatically determines the projection matrix to align the earbud local coordinate into the head coordinate system. For example, if our algorithm observes the best BCG responses along the Z, X, Y, respectively, the projection matrix will be following

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (1)$$

B. Breathing Depth Assessment

Breathing depth is an important metric to distinguish deep breathing exercises from shallow breathing. Breathing depth is an indicator of how much air is inhaled into the lung while someone is breathing. Inhaled air is proportional to lung expansion, and lung expansion is proportional to the breathing head motion. Since our algorithm is designed to capture the breathing head motion, the magnitude of the extracted breathing signal is considered as breathing depth.

Our breathing depth determination algorithms work in two modes: 1) shallow breathing detection for real-time feedback on breathing depth and 2) overall normalized breathing depth in percentage(%) to be reported at the end of the exercise. Fig. 4 shows the algorithm steps for both the earbud and the phone. We explain the algorithms for both cases in detail below.

1) *NonBreathing Head Motion Detection*: Chest expansion and contraction due to breathing moves the shoulder and the head. However, we often move our heads for many other purposes, for example, moving our heads from left to right. Nonbreathing head motion is a challenge when algorithms use earbud motion sensor data to track breathing. We have collected various nonbreathing head movements (e.g., slow and fast head movements, posture change, and random movements) to distinguish the nonbreathing head motion from the breathing head motion. We observe that the accelerometer magnitude range and the accelerometer lateral medial axis (Fig. 2) range are the most important feature computed on the 10 s window to distinguish nonbreathing head motion from breathing. We have trained thresholds for those two motion features optimizing the separation between the feature

distribution for breathing and nonbreathing head motion. We determine nonbreathing head motion if either of them is above the threshold. We make our algorithm sensitive to nonbreathing head motion since this is one of the biggest confounding factors for the earbud motion sensor-based breathing exercise tracking. If a nonbreathing head motion is detected, the earbud will send the user notification on the phone and discard the current window from further processing.

2) *Shallow Breathing Detection*: Our algorithm extracts the breathing waveform by removing the high-frequency components with a 5-sample median filter. We then compute the breathing depth features from the accelerometer data based on a 10s window to create depth biofeedback. We consider a 10 s window since the maximum inhalation or exhalation duration in our breathing exercises is 10 s to support slow-paced up to as slow as three BPM breathing exercises. Therefore, it is expected that this window would capture at least one inhalation or one exhalation phase to determine depth of breathing. Depth features include min, max, range, percentile range (difference between the 90th percentile and 10th percentile), and variance on each accelerometer axis and the accelerometer magnitude ($\sqrt{x^2 + y^2 + z^2}$). We have extracted 21 depth-related motion features from the accelerometer and gyroscope. We then determine each feature's importance based on information gain (Gini impurity [70]) measures in distinguishing shallow breathing from deep breathing exercises. We used Gini impurity-based feature importance. This metric calculates the degree of misclassification or uncertainty in a node or leaf of a decision tree model. A higher Gini impurity score indicates greater heterogeneity among instances in the node, suggesting that splitting on that particular feature might improve the classification performance. Feature importance derived from Gini Impurity represent the relative reduction in impurity achieved by each feature across all splits in the tree. This is simple to understand, computationally efficient, widely supported by popular libraries like Scikit-learn [71].

For instant feedback on whether the user's breathing is shallow or deep, we trained a classifier to distinguish shallow from deep breathing. In our dataset, we labeled regular breathing as shallow breathing and heavy and guided controlled breathing datasets as the deep breathing categories. We have 2179 rows in the training dataset, where around 45% of them are shallow breathing. We then train a Random Forest classifier using leave-one-subject-out (LOSO) validation for a generic shallow breathing detector model. While training the Random Forest model, we use 100 estimators with maximum tree depth 4 and bootstrap is set to *True* so that each tree utilizes a subsample of the dataset.

3) *Overall Normalized Breathing Depth Estimation*: Finally, for the end-of-the-session breathing performance assessment, we keep buffering the depth features on the phone side to calculate user-interpretable normalized breathing depth in percentage(%). We have selected the percentile range of the superior inferior axis (AccelAmplitude) as the breathing depth-determining feature since this is one of the top features in distinguishing deep breathing from shallow breathing. We further calculate the accel amplitude max feature using the

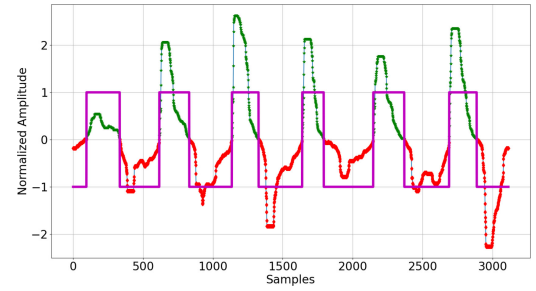


Fig. 5. Detecting the breathing phases using gyroscope data.

following equation:

$$\text{Accel Amplitude Max} = Q_3 + 1.5 * (Q_3 - Q_1) \quad (2)$$

where Q_1 , and Q_3 are the first and the third quartile of the inhalation amplitude derived from accelerometer data, respectively.

To make the depth information more insightful and actionable for the user, we normalize it with respect to the same feature extracted from the dataset consisting of the breathing exercises from 30 participants with varying ages, gender, and ethnicity (described in Section IV). Finally, the depth information is presented as a percentage (%) measure. It indicates how deep the breathing is in the current exercise compared to the maximum depth of breathing determined in the dataset

$$\text{Depth} = 100 * \frac{\text{Accel Amplitude Max}}{\lambda} \% \quad (3)$$

where λ is the normalization factor trained from the dataset.

C. Real-Time Breathing Phase Tracking

Previous works [20], [21], [34], [72] showed the feasibility of tracking passive and in some cases real-time breathing using earbuds [20] and smartwatch accelerometers [21]. Our algorithm can detect breathing phases in real-time using 3-axis gyroscope data. Gyroscope is believed to be more robust to nonbreathing motion artifacts in breathing data [34]. We have modified the algorithm presented in [20] for the gyroscope data. It works on segmenting the data in a 2 s window with 200-ms sliding, detecting the main bud, and then selecting the best axis that aligns more with the breathing based on the orientation of the device. Previous works [20], [21] on monitoring mindful breathing also used a similar window size. This algorithm determines whether the earbud is worn properly in the ear with the right orientation. The orientation of the earbuds and the inertial motion sensing unit is shown in Fig. 2.

This figure highlights that when the user is performing breathing exercises, rotational breathing head motion occurs around the lateral medial axis. Moreover, we analyze all the rotations across the three-axis and observe that our algorithm selects gyroscope lateral medial axis to produces the best accuracy for breathing phase detection among others. It then extracts breathing waveform by applying the median filter,

normalizing the signal using z -score normalization shown in the below equation

$$Z_i = \frac{S_i - \mu}{\sigma} \quad (4)$$

where S_i is the sample at i_{th} index, μ is the sample mean, and σ is the sample standard deviation.

We then apply the median filter again to extract the underlying breathing waveform. We then transform the breathing signal into a square waveform (also considered as the phase signal) by the following formula (Fig. 5):

$$\text{Phase}_t = \begin{cases} -1, & \text{if } S_t < 0 \\ +1, & \text{if } S_t > 0 \end{cases} \quad (5)$$

This algorithm considers consecutive '+1' labels as inhalation and consecutive '-1' labels as exhalation. Our algorithms determine the duration of each phase in each breathing cycle based on the timestamp of the detected breathing phases. Hence, the duration of the consecutive +1's is the inhalation duration, and -1's is the exhalation duration. Taking the first difference on this phase signal detects all the phase transitions - exhalation-to-inhalation transition will be '+2' and inhalation-to-exhalation transition will be '-2'. The timestamps of the phase transition points are later processed on the phone side to determine the breathing rate.

Breathing symmetry is calculated as the ratio between consecutive inhalation and exhalation duration. We calculate the breathing symmetry for each breathing cycle [20]. Finally, we take the median of all the breathing symmetry calculated in a breathing exercise session. Breathing symmetry is important to track since target breathing exercises can be symmetrical or asymmetrical. If the user targets asymmetrical breathing exercises (shorter inhale and longer exhale), the breathing symmetry will be lower than one. If the user targets symmetrical breathing exercises (inhale and exhale have the same duration), the breathing symmetry should be closer to one.

D. Breathing Rate Estimation

We estimate breathing rate in two modes - 1) real-time breathing rate to guide the user to adjust their breathing pace and 2) end-of-the-session breathing rate to assess the performance of the current breathing exercise. In the first case, the breathing rate can be estimated based on the detected phase transition described in the previous section (Section V-C). If the current pace is deviated more than 2 BPM compared to the targeted pace, the phone UI will show a notification to adjust the user's breathing.

On the other hand, the phone can maintain a buffer of all phase transition labels with timestamps and performs filtering to smooth out falsely detected transitions. The final breathing rate (BR) is calculated from the median (Med) inhalation (In_{dur}) and exhalation (Ex_{dur}) duration

$$BR = \frac{60}{Med(In_{dur}) + Med(Ex_{dur})} \quad (6)$$

Finally, our algorithm determines the breathing rate more accurately at the end of the breathing session and compares the target and the estimated rate in the user performance

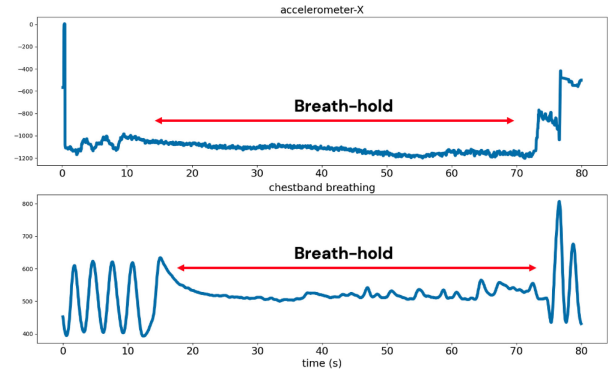


Fig. 6. Breath-hold signal in accelerometer superior inferior axis (top) and corresponding reference signal in the chestband.

visualization interface. The user thus can reflect on both their target and actual breathing rate and take informed decisions based on adjusting their target breathing rate to improve breathing performance.

E. Breath-Hold Detection

Breathing exercises with equal phases (same inhalation and exhalation duration) usually do not have breath-hold phase.

However, several mindful breathing exercises have breath-hold and they can help improve lung capacity [73]. Therefore, breath-hold is an important phase to detect. It can help recommend comfortable and effective breathing exercises based on the user's current health and fitness.

There are several challenges in detecting the breath-hold using earbud motion sensors. For example, breath-hold means there is no active breathing motion. However, the earbud motion sensor can still capture the ballistic motion of the heart pumping the blood. It is because heart motion is autonomous and cannot be controlled voluntarily like pausing the breath. Another challenge is to handle the baseline shift in the motion sensor data which sometimes can falsely look like a breathing (Fig. 6). Our algorithm handles those challenges. Breath-hold detection steps are described below.

1) *Preprocessing*: We used six-axis motion sensor data for breath-hold detection. Our preprocessing step includes smoothing each axis using a median filter, computing the accelerometer magnitude $\sqrt{x^2 + y^2 + z^2}$. We take the first difference of the signals to handle the baseline shift similar to motion-sensor based apnea detection algorithms [72], then segmenting the original axes and the magnitude signal into 2 s window. We have considered 2 s window to tradeoff between the accuracy and the fact that breath-hold in breathing exercises can last only a few seconds.

2) *Breath-Hold Feature Extraction*: We then extract features related to the breath-hold detection. It includes the percentile range (difference between the 90th percentile and 10th percentile to reduce the effect of the noise spikes in the signal) to capture the amplitude related information and standard deviation related to the variation of the signal. Features are extracted from each axis and the magnitude signal. We have extracted 12 features.

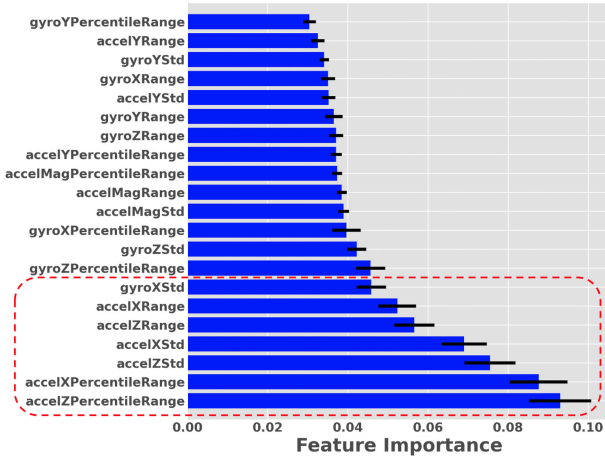


Fig. 7. Gini feature importance (the higher the more important) for shallow versus deep breathing classification. Top seven selected features are marked in a dotted red box. Percentile range is calculated as difference between percentile of (90th and 10th).

3) *Distinguish Breath-Hold and Breathing*: We train a classifier where we assign the positive labels with the features extracted from the annotated breath-hold segments (Fig. 6) and negative labels with the feature extracted from the regular breathing, deep breathing, and nonbreathing head-motion segments. We have 5271 rows in the dataset with balanced positive (49%) and negative classes (51%). We then train a Random Forest classifier using LOSO cross-validation to develop a generic classifier for detecting breath-hold.

4) *Breathing Rate With Breath-Hold Detection*: Once our algorithm detects the presence of breath-hold (Bh_{dur}) in a breathing cycle, the following formula estimates the breathing rate (BR):

$$BR = \frac{60}{\text{Med}(In_{dur}) + \text{Med}(Bh_{dur}) + \text{Med}(Ex_{dur})} \quad (7)$$

VI. EVALUATION

In this section, we describe the performance of each biomarker for tracking breathing exercises.

A. Shallow Breathing and Nonbreathing Head Motion Detection Accuracy

The shallow breathing classification accuracy is 72.62%, precision is 77.80%, recall is 75.46%, and F1-score is 74.27% based on the 10 s window. We also analyze the importance of the features to distinguish between shallow and deep breathing using Gini impurity and information gain. Range and standard deviation of the superior inferior axis and lateral medial axis features appear as the most important features (Fig. 7). We also observe that more accelerometer features are selected compared to gyroscope features. It is because breathing exercises involve more linear motion than that rotation motion. The lower accuracy can be explained by the fact that we consider regular breathing in our dataset as shallow breathing, which can be affected by self-consciousness. Conscious breathing can be closer to deeper breathing for some participants. We have also performed experiments by varying the window size

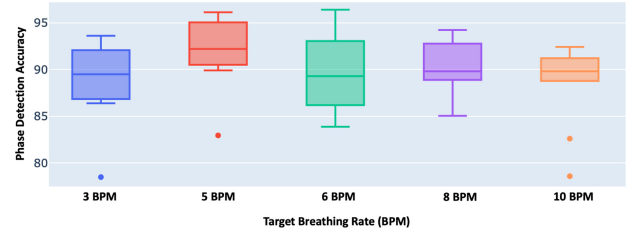


Fig. 8. Phase detection accuracy blackbox plots across different slow-paced breathing. BPM stands for breath per minute.

from 1 to 15 s. It reveals that the F1-score drops for window sizes less than 10 s and also does not improve it further for bigger window sizes. Hence, we found 10 s as the optimal window for shallow breathing detection and user notification. The head motion detection algorithm identifies nonbreathing head motion from the breathing motion with 83.87% accuracy.

We can also consider how many 10 s windows in a minute are detected as shallow or affected by nonbreathing head motion to filter out the data for the overall breathing performance calculation. In this way, our algorithms become more robust to various artifacts, including motion artifacts and reliable data can be presented to the user to reflect at the end of the exercise session.

B. Breathing Phase Detection Accuracy

Our algorithm can detect breathing phases (inhalation and exhalation) with accuracy 89.5%, precision 85.29%, recall 90.26%, and F1-score 87.71%. It is to note that accuracy is a more appropriate performance metric for breathing phase detection because it includes both the correct classification of inhalation and exhalation phases. Accuracy can be affected by motion artifact and shallow breathing to some extent. However, our system can still maintain the same 89.5% accuracy by detecting and filtering out the data affected by shallow breathing (two 10 s shallow breathing segments in a minute) and head motion (two 10 s nonbreathing head motion segments in a minute) using our algorithms. It shows that our algorithms are resilient to confounding artifacts.

We have further tested the algorithm across different slow-paced breathing exercises. It shows that the accuracy is similar across different breathing paces. We notice that the algorithm is performing lower in the 3 BPM case since this is a more challenging exercise for most of the participants (Fig. 8).

We also analyzed the test data collected using the demo breathing exercise tracking and real-time feedback system using Galaxy Buds2 and an Android smartphone (Section IV has more detail). We observed that 16 out of 30 subjects' breathing was shallow or affected by nonbreathing head motion. We have analyzed the accuracy for the remaining individual participants and compared the performance of our gyroscope-based breathing phase detection algorithm with the accelerometer-based algorithm presented in BreatheBuddy by Rahman et al. [20]. We have chosen BreatheBuddy [20] for comparison since this is the latest published work that uses an earbud motion sensor for real-time breathing phase detection. We observe that our algorithm based on gyroscope data shows similar accuracy as shown in Fig. 9.

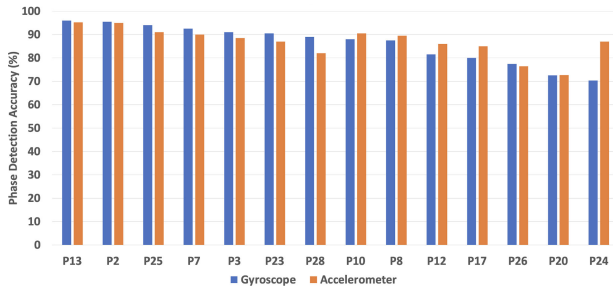


Fig. 9. Phase detection accuracy across different subjects comparing the gyroscope and accelerometer-based detection.

Moreover, our algorithm can consistently estimate Inhalation-Exhalation (IE) ratio with 0.32 overall MAE across all slow-paced breathing tasks, which is 29.35% of the overall ground truth of symmetry 1.09.

C. Breathing Rate Estimation Accuracy

We evaluate the breathing rate estimation accuracy in MAE. MAE is the most commonly used performance metric reported in previous research [34], [55], [74], [75]. Additionally, standard guideline from the Consumer Technology Association (CTA) [59] and U.S. Army Research Institute of Environmental Medicine (USARIEM) [76] use MAE as the performance evaluation criterion for breathing rate estimation. MAE is defined as the average of mean absolute differences between BR_{gt} and BR_{est}

$$MAE = \frac{\sum_{i=1}^n |BR_{gt_i} - BR_{est_i}|}{n}$$

Here BR_{est} is the estimated breathing rate.

Overall, MAE for deep breathing exercises is 0.61 ± 0.89 BPM (median absolute error 0.24 BPM). Our breathing estimation error is significantly lower than the MAE performance criteria set by the CTA respiratory monitoring standard [59]. Moreover, we observe root mean square error (RMSE) as 1.03 BPM. We also find that the mean absolute percentage error (MAPE) for breath rate estimation is $9.61\% \pm 14.75\%$. Since mindful breathing exercises are usually slow-paced breathing, small estimation errors in BPM can yield a higher number of percentage errors.

Fig. 10 shows the Bland-Altman plot to visualize the agreement between the estimated breathing rate and the reference breathing rate. For each pair of measurements, it shows the mean of the two against the difference between the two. It is a widely used tool to visualize fixed bias (i.e., a nonzero mean difference), proportional bias (i.e., nonzero slope for the line of best fit), and limits of agreement (i.e., between which two values of the y-axis do 95% data points lie). We observe a fixed bias of 0.46 BPM, a proportional bias of around 0, and limits of agreement between -1.4 and 2.3 . This is a fairly low fixed bias and negligible proportional bias compared to the existing works [26], [34], [35]. If desired, the fixed bias can be removed by subtracting 0.19 from all our predicted values. The negligible proportional bias suggests that the error in our prediction is not correlated with the magnitude of the predicted value.

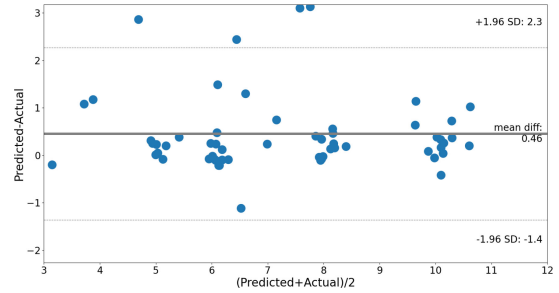


Fig. 10. Bland-Altman plot for breathing rate estimation performance. It shows that the estimation is distributed around zero. The MAE is 0.61 BPM (median: 0.24 BPM) with root mean squared error (RMSE) of 1.03 BPM.

Moreover, we also analyze robustness of our algorithms when motion artifacts and shallow breathing segments are present in a breathing exercise session. We examine how the number of nonbreathing head motion and shallow breathing segments in an exercise window affect the breathing rate estimation. We estimate the impact of motion artifact on breathing rate estimation accuracy in a minute long window. Utilizing the 10s window based nonbreathing head motion detection (as described in Section V-B1), we gradually admit 10, 20, 30, 40, 50, 60 s motion artifact during breathing exercise for this experiment. We observe that the more we admit the motion artifact, the higher the estimation error. However, overall estimation error is still very low (MAE 1.1 BPM) showing the evidence of the robustness of our algorithms in presence of motion artifacts. We also observe that the effect of the presence of shallow breathing is even lower than that of the nonbreathing head motion artifact.

D. Breath-Hold Detection Performance

Our classifier can detect breath-hold with accuracy 95.02%, precision 72.34%, recall 98.81%, F1-score 71.72% using LOSO cross-validation. Fig. 11 shows the feature importance in distinguishing breath-hold from the breathing signal. It shows that accelerometer magnitude range, percentile range, and standard deviation are the most important features in detecting breath-hold. It is to note that accelerometer magnitude features are robust to earbud orientation. It means that the classifier can detect breath-holds independent of the earbud placement inside the ear.

We further analyze the breath-hold detection performance across different window sizes. We observe that 4-s window size shows the best precision 87.23% and F1-score 84.79% along with accuracy 94.67% and recall 95.97% using LOSO cross validation. However, this model may only work for the breathing exercises having at least 4-s breath-hold in between the inhalation and exhalation phases.

VII. USE-CASE FOR BREATHING EXERCISE BIOFEEDBACK USING EARBUD BIOMARKERS

Comprehensive breathing biomarkers captured by our MindfulBuddy systems can be used to design numerous UI and user experience. We showcase two examples for the most common user scenarios for breathing exercises - first, breathing

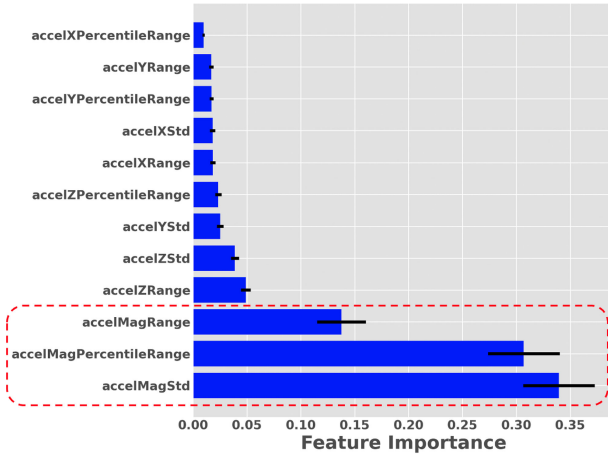


Fig. 11. Gini feature importance for breath-hold detection. Top three selected features are marked in a dotted red box. Accelerometer magnitude ($\sqrt{x^2 + y^2 + z^2}$) range, accelerometer magnitude percentile range, accelerometer magnitude standard deviation are the three most important features to detect breath-holding.

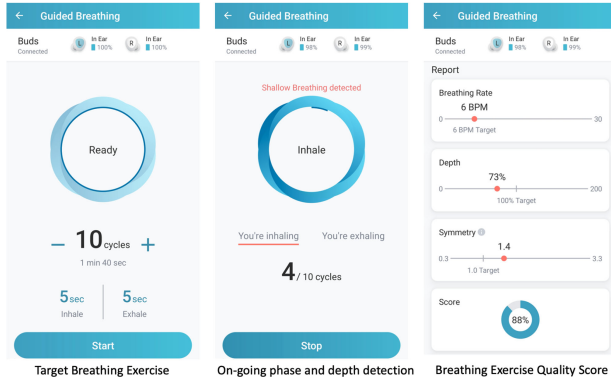


Fig. 12. Breathing exercise app interface. User can select a target breathing exercise, then follow the guidance while our algorithm detect on-going phases, head motion, and breathing depth. At the end, the interface shows the breathing quality scores for each breathing biomarker and the composite breathing score at the bottom.

biofeedback for guided breathing exercises with a breathing performance score, and second, breathing biofeedback for eye-closed meditation.

A. Breathing Score for the Biofeedback on Performance

Breathing performance can be defined as how closely the subject performs the breathing exercise compared to the target breathing pattern. The most commonly used breathing metric is the breathing rate measured in BPM which becomes a common health metric of consumer wearables, such as smartwatches [77], [78], [79].

We propose a method to determine the breathing score based on how the user follows the breathing guidance (Fig. 12). The breathing score is designed to intuitively give more score to the user who can better follow the guidance. We hypothesize that it should help keep users engaged in breathing exercises and perform deeper breathing. We propose the following formula to combine the most commonly used metric (breathing rate) and the most user-perceived breathing parameter (breathing

depth)

$$\text{Score}_{BR} = 100 * e^{\frac{-\Delta}{3}} \quad (8)$$

$$\text{Score}_{BD} = 1.33 * \max(\min(D, 100) - 24, 0) \quad (9)$$

$$\text{Score}(\%) = \alpha * \text{Score}_{BR} + (1 - \alpha) * \text{Score}_{BD} \quad (10)$$

where S_i is the sample at i th index, D is the breathing depth, μ is the sample mean, σ is the sample standard deviation, Δ is the absolute difference between the target breathing rate and the actual breathing rate.

Equation (8) calculates the part of the score for the breathing rate. If the user maintains the overall target breathing rate throughout the exercise, they get full credit. The score exponentially diminishes if they do not comply with the guidance. The formula to calculate the contributions from the breathing depth is given in Equation (9). It does not give credit on depth score, if the breathing depth is very low ($< 25\%$) and gives full credit if it is above 100% . The higher the breathing depth, the better the score. Here, our composite score uses the constant $\alpha = 0.65$ for calculating the weighted average of the breathing score based on the breathing rate and breathing depth score. We give more emphasis to the users for meeting the breathing rate target, since it is more interpretable and clearer to follow the breathing guidance on the app. Final score is calculated by combining the scores from breathing rate and breathing depth as shown in Equation (10).

B. Qualitative Performance - Breathing Tracking System With Biofeedback and Breathing Score

At the end of the study, participants were given the SUS [64] to investigate if they found the breathing exercise interface helpful in monitoring their breathing exercises. SUS is a widely 10-item scale that helps evaluate the usability of an interface [64]. We used the participants' responses ($n = 25$) for the SUS questionnaire to compute the mean, as suggested in [64]. Due to incomplete responses, we discarded responses from 5 participants. The mean SUS score of our system was 76.5. A SUS score above 68 is considered above average in terms of usability [80]. Hence, the breathing exercise system was above average in terms of usability.

To evaluate if the Breathing score for the breathing task was helpful, at the end of the user study, we asked participants to respond to this 5-point Likert question: "The performance score was helpful to understand my breathing performance." The options for this question were: 1) strongly disagree; 2) disagree; 3) neutral; 4) agree; and 5) strongly agree.

Two participants selected strongly agreed, 15 participants agreed, five participants shared neutral sentiments toward the breathing quality score, and eight participants disagreed that the breathing quality score was helpful for them to understand their breathing performance. Overall, the majority of the participants ($N = 17$) found the composite score helpful (Fig. 13).

C. Meditation Biofeedback Without Guidance

Users may prefer to focus on meditation with their eyes closed and not looking into the phone screen for the breathing

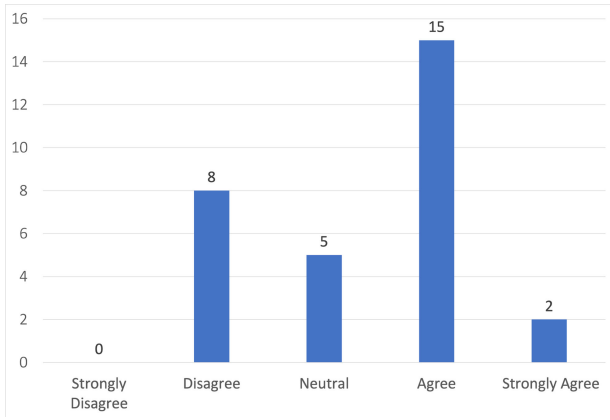


Fig. 13. Participants' responses toward the helpfulness of the breathing score for interpreting the breathing performance.

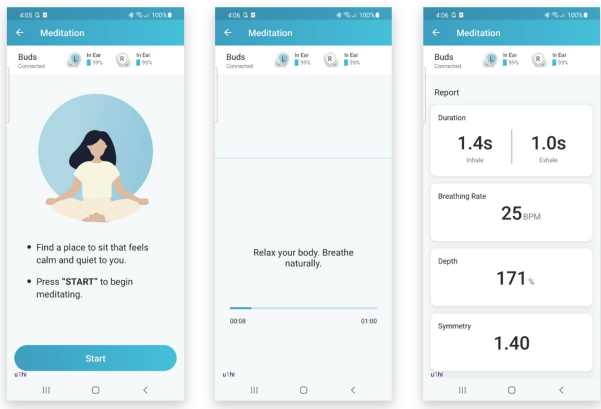


Fig. 14. Breathing exercise demo app interface for meditation. User can perform the closed-eye meditative breathing exercises without guidance and can still get the breathing biomarkers at the end of the breathing exercise session as a biofeedback.

guidance. Our breathing biomarker algorithms can support this usecase by passively tracking the breathing. We have tested our algorithm for this usecase and observe that our algorithm can estimate breathing rate of a meditative breathing exercise with similar accuracy. Fig. 14 shows an example UI design for meditative app that can passively track the breathing using the connected earbud and provide the biofeedback for the user.

VIII. SYSTEM IMPLEMENTATION

One of the main objectives of our algorithm design was to implement them in earbuds. Current earbuds have limited resources, so we focus our algorithms on being lightweight. To determine the feasibility of the on-device implementation, we fully implemented the algorithm pipeline on a commercially available Samsung earbud (Galaxy Buds2). We implement all the components of the signal processing algorithm in C to run the algorithm on the microcontroller. We also develop a companion smartphone application to communicate with the earbud via Bluetooth low-energy (LE) and show the guidance and the feedback on the ongoing breathing phase, head movement status, and shallow breathing status continuously in real-time. The algorithm takes 4.6 KB of earbud device memory.

Power consumption. Inertial motion (IMU) sensors, such as accelerometers are low-power sensors compared to other commonly used earbud sensors, such as microphones. IMU only consumes 0.5 mA power in the high-performance mode, which is 20 times lower than the microphones. “Always-on” sensing mode consumes even lower power. Moreover, it is now standard practice for commercial earbuds to use low-power co-processors called ‘sensor-hubs’ for reading, buffering, and processing continuous sensor data [81].

Furthermore, mindful breathing exercises last only a few seconds [20] to a few minutes [22]. We observe in our experiment that a 1-min breathing exercise session consumes only 0.05% earbud battery. In other words, only 1% battery can facilitate 20 one-minute breathing exercise sessions. Usually, people perform breathing exercises on-demand or 2–3 times per week. Under this assumption, the user can perform several weeks of breathing exercises by consuming only 1% battery.

Latency. Our algorithms in the earbuds take around 0.04 s time to execute. The algorithm outputs are shared with the connected Android phone with bluetooth low energy and this communication may take up to 0.5 s based on the Android operating system's priorities. When the subject continues to perform a particular breathing phase (e.g., exhalation) for a longer period of time, they may not feel the communication delay on the phone interface unless there is a transition from one phase to the other. Moreover, the UI and interaction can be designed in such a way so that the delay can be abstracted from the user and they can still feel the smooth phase transitions.

IX. DISCUSSION, LIMITATIONS, AND FUTURE WORKS

In this section, we discuss the implications of our approach in terms of various real-life contexts, describe the limitations, and explain how future works can address such limitations.

A. Algorithm Yield and Coverage Improvement

Shallow breathing and nonbreathing head motion make the algorithm harder to accurately estimate breathing biomarkers using earbud motion sensors. In the worst case scenarios, the system won't be able to show any feedback based on the user's actual breathing. We can handle this case and improve the yield and coverage (retention rate [26]) by complementing the audio-based breathing detection. Recent works [26] show that breathing sound can be detected using earbud audio. Particularly, Ahmed et al. [26] showed that the data yield and coverage can be improved for monitoring regular breathing when both motion sensor and audio from earbuds. Usually, breathing sound from mindful breathing exercises are more audible than the regular breathing sounds [26], [36]. We have applied the breathing phase transition detection algorithm presented in [26] on our breathing exercise audio dataset. We observe that algorithm yield can be improved 14% on average when we complement the audio with the motion sensor data for breathing exercise tracking. Furthermore, it improves the breathing rate estimation accuracy with only 0.56 BPM MAE and RMSE 1.01 BPM.

B. Posture and Mindful Breathing Exercises Using Earbuds

Breathing exercises can be practiced in different body postures. Sitting down cross-legged with the hands resting on our thighs gives us greater stability and has been used for thousands of years [82], [83], [84], [85]. Someone having issues with hip and knee joints causing difficulty in sitting cross-legged can do the exercise while sitting on a chair [84]. Hence, we focus more on sitting upright position and have collected most of the deep breathing exercises in sitting posture.

We further analyzed our study's limited deep breathing exercise sessions collected in lying posture. We observe that breathing motion captured by the earbud motion sensors in lying posture is much shallower than that of the sitting posture. It is because the head stays firm on the headrest, which limits the head motion, and the breathing motion is mostly demonstrated through thoracic and abdominal motion [65], [86]. Therefore, lying down posture is the worst-case scenario for the algorithms presented in this article. Our shallow breathing detector detects the breathing sessions in the lying posture as shallow breathing, creates shallow breathing notification (Fig. 1). Future works can explore posture the detection and customize the signal filtering algorithms for lying down posture so that breathing signals can be amplified for detecting them as deep breathing exercises.

C. Enabling Fine-Grained Real-Time Breathing Feedback

Our lightweight, low-power algorithms can passively offer fine-grained respiratory biomarkers for designers to design various mindful breathing-centric biofeedback systems and user experiences. We have implemented our algorithms in a demo application (Section VII), which provides real-time feedback on breathing depth (shallow or deep), ongoing breathing phase (inhale or exhale), whether nonbreathing head motion is present in the signal, inhalation/exhalation ratio, breathing depth, and breathing score. These biomarkers can be leveraged for designing interactive experiences.

For example, Shih et al. presented a design of a gamified biofeedback-guided visualization on a smartphone to facilitate slow-paced breathing training based on the detected breathing phases [22]. The authors represented the inhalation (one direction) and exhalation (opposite direction) as the wind moving a sailboat, and the sailed distance is represented as the phase duration. Though this is just one example of how information related to one's breathing biomarkers can be used for designing interactive biofeedback experiences for end users, it opens up many opportunities for designers, which is enabled by our biomarker algorithms.

D. Closed Loop System to Improve Mental Well-Being

Stress-induced disorders like hypertension and other cardiovascular diseases are fast-growing epidemics in modern society [87]. Recent research shows that earbuds can detect physiological responses due to psycho-social stress [88]. There are earbud-based consumer mental health monitoring products in the market that can passively monitor mental state [89]. Our proposed system can complement these existing technologies

to help users reduce stress, improve calmness, increase focus, and improve overall mental well-being all in one place. Moreover, recent studies [34], [35] show that earbud motion sensors can also passively monitor regular shallow breathing. Hence, it is possible to design an earbud system that can passively monitor breathing patterns and mental state for the user and recommend just-in-time adaptive (JITAI) [90] breathing exercises. Since the earbud is an increasingly popular mobile device, it can help users at work, at home, and during the commute.

X. CONCLUSION

In this article, we present a comprehensive but lightweight breathing exercise tracking system that monitors breathing biomarkers, including depth of breathing, breath-hold, inhalation phase, exhalation phase, IE ratio, and breathing rate using low-power earbud motion sensors. Our algorithms can handle nonbreathing head motion which is one of the biggest confounding factor for reliable breath monitoring using earbuds. Moreover, we present a novel physiologically informed earbud orientation handling algorithm by projecting the earbud axis into foot-to-head body axis. We believe that this system has a great potential to make the consumer earbud more useful to monitor and manage user's well-being toward healthy and happier life.

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